Day 2: Network Topologies

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Course: Diploma in Cyber Security

Topic: Day 2: Network Topologies

Introduction

Network topology describes how devices are arranged and connected. Common types include bus, star, ring and mesh. Each design has different cost, reliability and maintenance needs.

Task 1

Figure: Traditional Coaxial Bus Topology

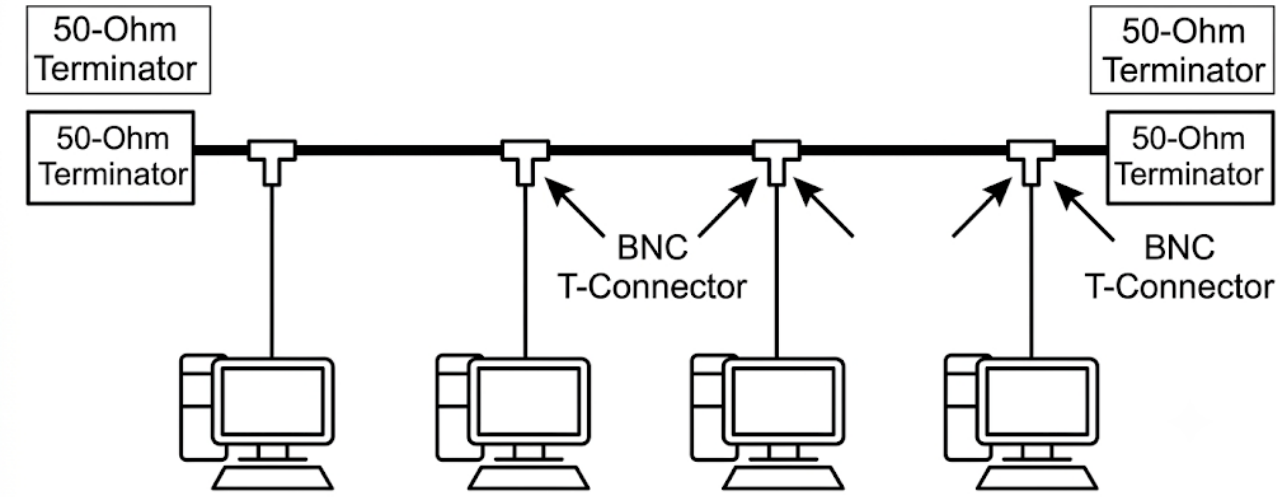


Figure: Traditional Bus Topology Using Coaxial Backbone Cable

Example: Older Ethernet networks using coaxial cable used bus-style layouts.

Task 2

PRACTICAL

Open Cisco Packet Tracer and create a simple star topology:
1. Add one switch.
2. Add four PCs.
3. Connect each PC directly to the switch using Ethernet connections.
4. Rename the PCs: WhatsApp, Zomato, Spotify and Flipkart.

5. Arrange the PCs around the switch.

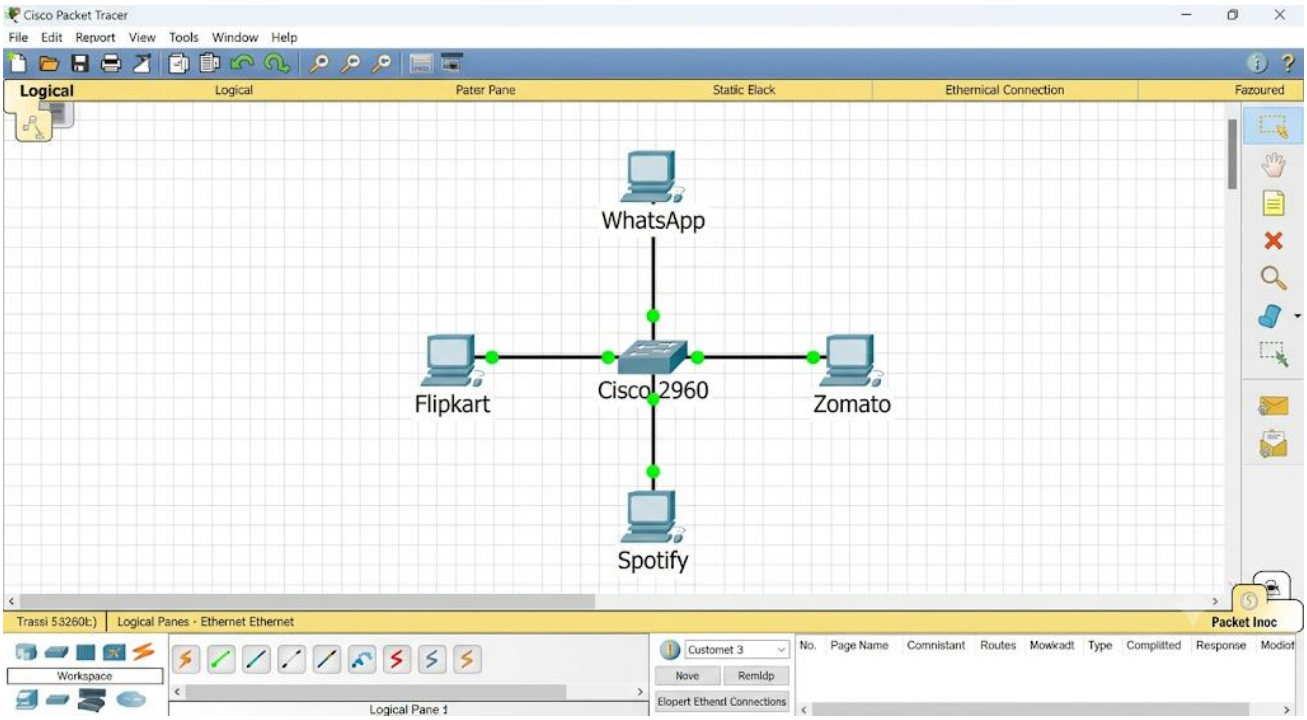


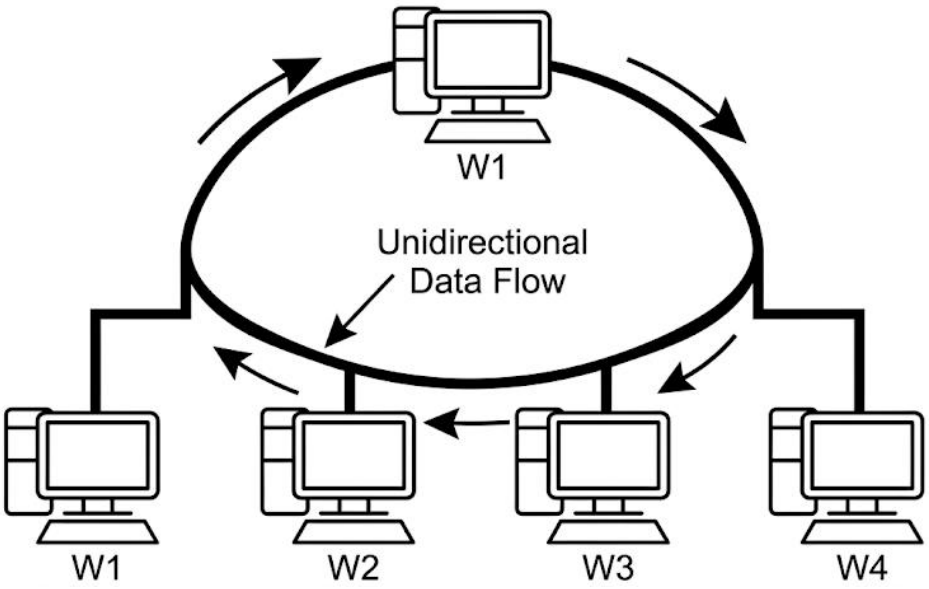
Figure: Cisco Packet Tracer Star Topology with Switch and Renamed Workstations

Configuration Summary:

- **Central Device:** Cisco 2960 24-Port FastEthernet Switch.
- **Connected Hosts:** 4 PCs directly connected via Copper Straight-Through cables (FastEthernet interfaces).
- **Configured Device Labels:** WhatsApp, Zomato, Spotify, and Flipkart.
- **Topology Type:** Star Topology (traffic between any two nodes must pass through the central switch).

Task 3

Figure: Unidirectional Ring Topology



Draw the devices in a circle, with each device connected to two neighbours. Add arrows in one direction around the ring. In a ring, data moves from one device to the next until it reaches the destination. In a simple logical ring, each device forwards the data along the ring.

Task 4

Topology	Advantages	Disadvantages
Mesh	High reliability; multiple paths	More cables/links; higher cost
Star	Easy to manage; one device failure usually affects only one link	Central switch failure affects many devices; needs more cable than bus

Conclusion

The best topology depends on the situation. Star is common because it is easy to manage, while mesh can provide high reliability but needs more connections. Bus and ring are useful for learning the basic ideas and are also seen in some older or specialised designs.

Quick Revision

- Star uses a central switch.
- Bus uses a shared backbone.
- Ring forms a loop.
- Mesh has multiple direct paths.
- Topology affects cost, reliability and troubleshooting.